# Data Visualization Using Tableau

## Aim

To learn the Tableau interface, connect to various data sources such as Excel, CSV, and SQL databases, create basic visualizations using bar charts, line charts, and pie charts, create calculated fields, and build interactive dashboards and stories for effective data analysis.

## Procedure

1. Install and open Tableau Public or Tableau Desktop.
2. Explore the Tableau interface, including the Data pane, Rows shelf, Columns shelf, Marks card, and worksheet area.
3. Connect to a data source such as Excel, CSV, or an SQL database.
4. Load the dataset and check the data types of the columns.
5. Create a bar chart to compare sales across different products.
6. Create a line chart to analyze sales trends over time.
7. Create a pie chart to visualize the proportion of sales by product.
8. Create calculated fields using Tableau formulas.
9. Combine multiple visualizations into a dashboard.
10. Create a story using story points to present the analysis step by step.

11. Format the charts with suitable titles, labels, legends, and colors.
12. Save or publish the workbook as required.

**Sample Dataset**

Create an Excel or CSV file named SalesData.csv with the following data.

| Date | Product | Region | Sales | Quantity |
|---|---|---|---|---|
| 01-01-2026 | Laptop | Chennai | 50000 | 2 |
| 05-01-2026 | Mobile | Madurai | 30000 | 3 |
| 10-02-2026 | Laptop | Chennai | 60000 | 2 |
| 15-02-2026 | Tablet | Coimbatore | 20000 | 2 |
| 20-03-2026 | Mobile | Madurai | 40000 | 4 |
| 25-03-2026 | Tablet | Chennai | 25000 | 1 |

**Connecting to Data Sources**

**A. Excel file**

1. Open Tableau.
2. Under Connect, select Microsoft Excel.
3. Choose SalesData.xlsx.
4. Select the required worksheet and open a worksheet to start visualizing the data.

**B. CSV file**

1. Under Connect, select Text File.
2. Choose SalesData.csv.
3. Check the data preview and confirm that the columns are correctly identified.
4. Open a worksheet.

**C. SQL database**

1. Under Connect, select the required database connector, such as Microsoft SQL Server or MySQL.
2. Enter the server name, database name, and authentication details.
3. Select the required table.
4. Load the data and open a worksheet.

**Creating Basic Visualizations**

**A. Bar chart**

1. Open a new worksheet and rename it Sales by Product.
2. Drag Product to Columns.
3. Drag Sales to Rows.
4. Select Show Me → Bar Chart if the chart is not displayed automatically.
5. Enable labels from the Marks card and add a suitable title.

Expected output: a bar chart comparing sales for Laptop, Mobile, and Tablet.

**B. Line chart**

1. Create a new worksheet named Sales Trend.
2. Drag Date to Columns.
3. Drag Sales to Rows.
4. Choose the Line mark type.
5. Select the appropriate date level, such as Month, and add the title Sales Trend Over Time.

Expected output: a line chart displaying sales changes over time.

**C. Pie chart**

1. Create a new worksheet named Sales Distribution.
2. Select Pie from the Marks card.
3. Drag Product to Color.
4. Drag Sales to Angle.
5. Drag Sales to Label to display the values.
6. Add the title Sales Distribution by Product.

Expected output: a pie chart showing each product's contribution to total sales.

**Creating Calculated Fields**

Calculated fields allow new values to be derived from existing data.

In Tableau, right-click in the Data pane and select Create Calculated Field.

**A. Total Amount**

Field name: Total Amount

[Sales] * [Quantity]

This calculates the product of sales and quantity for each row.

**B. Sales Category**

Field name: Sales Category

IF [Sales] >= 40000 THEN "High Sales"

ELSE "Low Sales"

END

This classifies each row as High Sales or Low Sales based on its Sales value.

**C. Total Sales**

Field name: Total Sales

SUM([Sales])

This expression calculates the total sales when used as an aggregate calculation.

# Building a Dashboard

## Sample Tableau Sales Dashboard

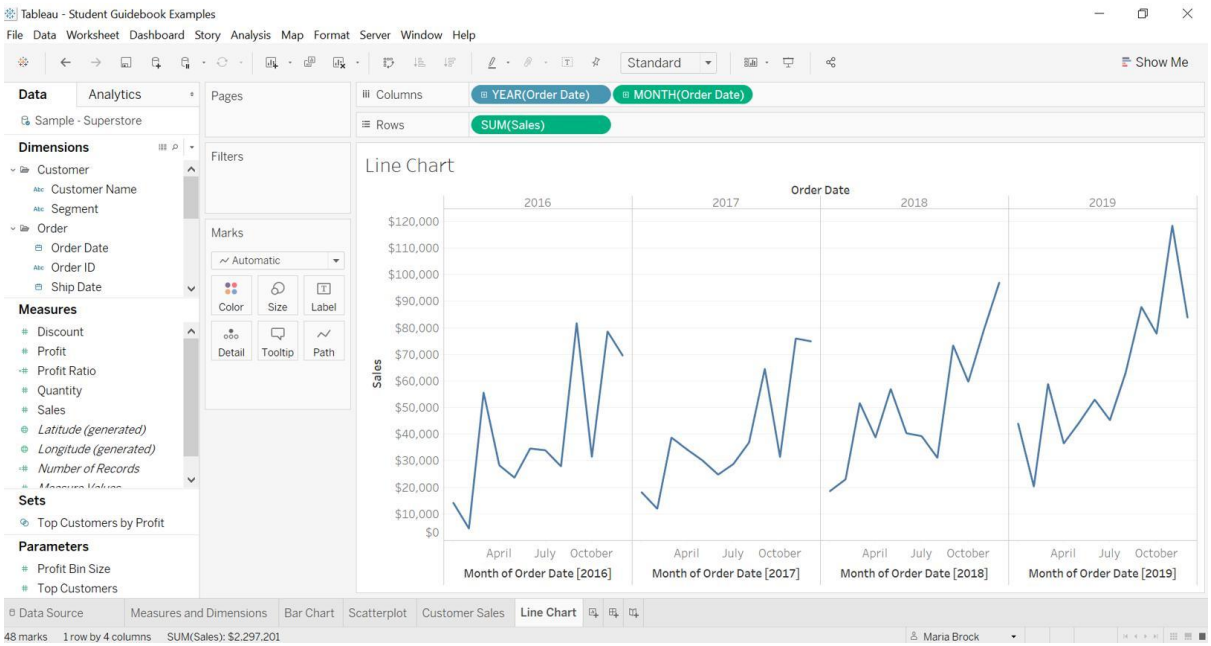
Total Sales

**₹2,25,000**

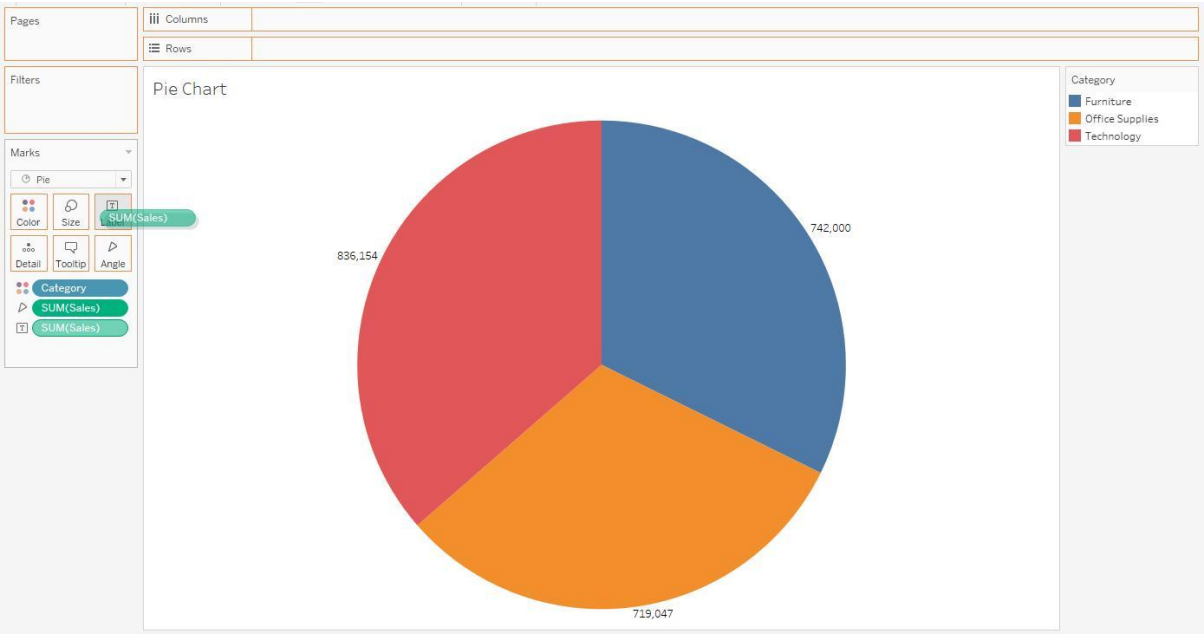
Total Quantity

**14**

Sales by Product




Sales Trend




Sales Distribution

Steps to create the dashboard:

1. Click the New Dashboard icon at the bottom of the Tableau window.
2. Drag the Sales by Product, Sales Trend, and Sales Distribution worksheets onto the dashboard.
3. Add a title such as Sales Analysis Dashboard.
4. Add a filter for Region to allow users to view sales by location.
5. Adjust the chart sizes and layout.
6. Save the workbook as SalesDashboard.twbx or the appropriate Tableau workbook format.

**Creating Stories in Tableau**

A Tableau story presents a sequence of visualizations or findings.

1. Click the New Story icon at the bottom of the workbook.
2. Select an appropriate story size.
3. Drag the Sales by Product worksheet onto the first story point.
4. Add a caption such as Product-wise Sales Comparison.
5. Add another story point containing the Sales Trend worksheet.
6. Add a third story point containing the sales dashboard.
7. Write a short caption for each story point to explain the finding.
8. Save the workbook.

**Output**

The expected output includes:

- A bar chart comparing product-wise sales.
- A line chart showing sales trends over time.
- A pie chart representing the distribution of sales.
- Calculated fields for total amount and sales category.
- A dashboard combining multiple worksheets.

- A story presenting the analysis through sequential story points.

**Result**

Thus, data visualization using Tableau was studied successfully. Data was connected from Excel, CSV, and SQL databases. Bar charts, line charts, and pie charts were created to analyze sales data. Calculated fields were developed to derive additional information, and dashboards and stories were built to present the analysis in an interactive and understandable manner.